Enterprise SLM Benchmarking: Local Feasibility to Cloud Strategy

📋 Executive Summary

This project presents a performance-driven evaluation of **Small Language Models (SLMs)**. By conducting rigorous benchmarking on local consumer hardware, I established a "performance floor" to architect a scalable, cost-optimized **Cloud Inference Strategy**. The study focuses on three models: **Llama 3.2-3B**, **Phi-3-Mini**, and **Mistral-7B**, evaluating them across five distinct cognitive tasks.

🏗️ Technical Stack & Methodology

- **Inference Engine:** Ollama (Local Environment)
- **Target Cloud Arch:** NVIDIA A100/H100 GPU Clusters / Serverless Inference
- **Quantization:** 4-bit (Q4_0)
- **Metrics: * Tokens Per Second (TPS):** Measures total generation throughput.
  - **Time to First Token (TTFT):** Measures responsiveness and system latency.

📊 Performance Benchmark Data

The following data was derived from local benchmarking to simulate resource-constrained environments and justify cloud migration for enterprise scale.

1. Throughput Analysis (TPS)

Task Type	Llama 3.2-3B	Phi-3-Mini	Mistral-7B
Reasoning	**15.6**	12.2	7.7
Code Generation	14.7	13.8	7.6
Summarization	13.2	13.2	7.3
Creative Writing	12.9	11.1	7.2
Factual QA	13.6	13.2	6.9

2. Responsiveness Analysis (TTFT in ms)

Lower values indicate better performance.

Task Type	Llama 3.2-3B	Phi-3-Mini	Mistral-7B

Summarization	2245.2	**2138.8**	2210.7
Code Generation	2264.4	**2142.1**	2201.0
Factual QA	2251.9	**2145.4**	2215.3
Creative Writing	2256.0	**2150.9**	2217.1
Reasoning	2265.4	**2143.3**	2207.7

💡 Key Findings & Strategic Insights

1. The "Parameter Tax"

Moving from a 3B parameter model to a 7B model (Mistral) resulted in a **~47% speed penalty** on local hardware. While Mistral offers deeper reasoning, its 7.34 TPS average identifies a critical memory bandwidth bottleneck that requires cloud-based GPU acceleration (A100/H100) for production-grade SLAs.

2. Responsiveness King: Phi-3-Mini

Phi-3-Mini consistently delivered the lowest TTFT across all tasks, averaging ~2144ms. For real-time user interfaces where "instant" feedback is critical, this 100ms+ advantage over Llama 3.2 makes Phi-3 the superior choice for frontend-integrated assistants.

3. Throughput Leader: Llama 3.2-3B

With a peak of 15.6 TPS in reasoning tasks, Llama 3.2-3B is the most efficient choice for high-volume batch processing, such as document summarization or automated data extraction.

🚀 Cloud Migration & Recommendations

Based on the local performance floor, the following cloud strategy is recommended for enterprise deployment:

- **Horizontal Scaling:** Deploy Llama 3.2-3B on a load-balanced cluster of small GPU nodes to maintain sub-second response times for high-concurrency summarization tasks.
- **GPU Allocation:** Transition Mistral-7B to dedicated NVIDIA instances to overcome the 7.34 TPS bottleneck and achieve fluid, real-time reasoning.
- **Hybrid Deployment:** Utilize Phi-3-Mini for low-latency edge intent classification, failing over to Cloud-hosted Mistral-7B only for high-complexity edge cases.

📂 Repository Contents

- `benchmark_data.csv` : Raw metrics for all model/task combinations.
- `visualization_dashboard.jsx` : Interactive React-based performance dashboard.
- `summary_report.md` : Technical analysis and conclusion.

Created as part of a technical feasibility study for scalable AI architecture.